HireTrack: Project Overview & Technical Architecture

Mission: To give hiring teams a self-hosted applicant tracking system that covers the full recruitment loop — jobs, candidates, interviews, and communication — without forcing the organization onto a SaaS vendor's database.

The Core Problem and The Solution

Most teams either drown in spreadsheets or rent an ATS they cannot control. Spreadsheets have no pipeline, no audit trail, and no real-time collaboration. Hosted ATS products lock candidate data in someone else's cloud, charge per seat, and resist the custom fields, lanes, and branding that agencies and in-house HR actually use.

HireTrack is a multi-tenant product you run on your own server. Each organization gets its own workspace, career portal, roles, and settings. HR moves candidates on a Kanban board; interviewers see only what they are assigned; candidates apply through a branded public portal. AI scoring, email, calendars, and file storage plug in as organization settings rather than a separate product.

System Architecture & Runtime Workflow

HireTrack is a single Next.js application with a custom Node HTTP server. The UI and API share one process. Real-time updates do not go through a separate realtime vendor.

1. **Identity and tenancy.** Users belong to an Organization. JWT sessions and role-based access (Administrator, HR, Interviewer, Candidate) scope every query. Organizations cannot read each other's jobs or candidates.
2. **Jobs and the career portal.** HR publishes jobs with custom application fields. Candidates apply at a public branded portal (*apply-jobs*), including guest apply. Files land on local disk, Cloudinary, or AWS S3 according to org settings.
3. **Pipeline.** Candidates sit in lanes (default or per-job). Drag-and-drop, checklists, remarks, and a history collection form the audit trail. Search and filters run against MongoDB, including custom fields.
4. **Realtime.** Production starts via `server-ws.cjs`: Next.js plus a WebSocket server on the same port. Org-scoped broadcasts move cards, counts, and notifications to connected dashboards. SSE remains a fallback path for older clients.
5. **AI and documents.** Google Gemini parses and scores resumes (skills, experience, affordability, overall). Bulk CV upload can OCR scanned PDFs with Tesseract and an optional local markdown-parser sidecar.
6. **Communication.** Templated email (SMTP, Resend, or Brevo). Interviews can create Google Meet, Outlook, Microsoft Teams, or Zoom events. Reminders run from cron on the host.
7. **Operations.** A customer does not clone and `next build` on the VM. They run `installer.sh`, which registers the license, installs MongoDB or points at Atlas, nginx, and PM2, then pulls a pre-built release tarball. In-app updates call the same installer.

Target Audience Positioning

- **In-house HR and TA teams** that want a full pipeline (jobs → screen → interview → offer) with branding on the career page.
- **Recruitment agencies** that need multiple organizations, custom lanes, and high candidate volume without per-seat SaaS pricing.

- **Teams that must self-host** (data residency, intranet, or air-gapped networks) while still receiving versioned builds from the vendor.

Technology Stack Summary

Domain	Technology	Implementation Purpose
Application	Next.js (App Router), TypeScript, React	Pages, dashboard UI, and HTTP API in one codebase.
Client state	Redux Toolkit, RTK Query	Auth/org persistence and typed API access. No ad-hoc fetch layer.
UI	Tailwind CSS, Radix / shadcn-style components	Accessible forms, dialogs, Kanban, calendar.
Database	MongoDB, Mongoose	Organizations, users, jobs, candidates, history, settings, notifications.
Realtime	ws (WebSocket) on `server-ws.cjs`; SSE fallback	Live pipeline and notification updates without polling every client.
AI	Google Gemini (optional Groq / other keys in settings)	Resume extraction and multi-axis candidate scoring.
Files & mail	Local disk, Cloudinary, AWS S3; SMTP / Resend / Brevo	CVs, attachments, and transactional email.
Process & edge	PM2, nginx, Let’s Encrypt	Process supervision, TLS, reverse proxy on customer VMs.

Licensing (License Admin) and the release/installer pipeline are separate services that sit around this application. They are documented in companion overviews.